

PF (PROGRAMMING FUNDAMENTAL)

ASSINGNMENT#03:

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Section – 2C

PROBLEM#01:

Average in an Array

```
#include <iostream>
```

```
#include <iomanip>
```

```
using namespace std;
```

```
double calculateAverage(int arr[], int size)
```

```
{
```

```
    int sum = 0;
```

```
    for (int i = 0; i < size; i++)
```

```
    {
```

```
        sum += arr[i];
```

```
    }
```

```
    return (double)sum / size;
```

```
}
```

```

int main ()
{
    int n;
    cin >> n;

    int scores[n];

    for (int i = 0; i < n; i++)
    {
        cin >> scores[i];
    }

    cout << fixed << setprecision(2);
    cout << "Average Score: " << calculateAverage(scores, n);

    return 0;
}

```

PROBLEM#02:

Largest and Smallest value

```

#include <iostream>
using namespace std;

```

```

int calculateLowest(int arr[], int size)

```

```
{  
    int lowest = arr [0];  
  
    for (int i = 1; i < size; i++)  
    {  
        if (arr[i] < lowest)  
            lowest = arr[i];  
    }  
  
    return lowest;  
}
```

```
int calculateHighest(int arr[], int size)  
{  
    int highest = arr[0];  
  
    for (int i = 1; i < size; i++)  
    {  
        if (arr[i] > highest)  
            highest = arr[i];  
    }  
  
    return highest;  
}
```

```

int main ()
{
    int n;
    cin >> n;

    int scores[n];

    for (int i = 0; i < n; i++)
    {
        cin >> scores[i];
    }

    cout << "Lowest Score: " << calculateLowest(scores, n) << endl;
    cout << "Highest Score: " << calculateHighest(scores, n);

    return 0;
}

```

PROBLEM#03:

Custom Power Function

```

#include <iostream>

using namespace std;

int calculatePower(int base, int exponent)

```

```

{
    int result = 1;

    for (int i = 0; i < exponent; i++)
    {
        result *= base;
    }

    return result;
}

int main ()
{
    int base, exponent;

    cin >> base >> exponent;

    cout << base << " raised to the power "
        << exponent << " is: "
        << calculatePower(base, exponent);

    return 0;
}

```

PROMBLEM#04:

Voltage Calculate

```
#include <iostream>

using namespace std;

double calculateVoltage(double current, double resistance)
{
    return current * resistance;
}

int main ()
{
    double current, resistance;

    cin >> current >> resistance;

    cout << "Total Voltage: "
         << calculateVoltage(current, resistance)
         << " V";

    return 0;
}
```

PROBLEM#05:

Cricket Team Initial Finder

```
#include <iostream>
```

```
using namespace std;
```

```
bool linearSearch(char arr[], int size, char target)
```

```
{
```

```
    for (int i = 0; i < size; i++)
```

```
    {
```

```
        if (arr[i] == target)
```

```
            return true;
```

```
    }
```

```
    return false;
```

```
}
```

```
int main ()
```

```
{
```

```
    char team [11] = {'b','r','s','f','h','i','a','m','n','k','z'};
```

```
    char target;
```

```
    cin >> target;
```

```
    if (linearSearch(team, 11, target))
```

```
        cout << "Player is in the team";
```

```
else  
    cout << "Player not found";  
  
    return 0;  
}
```

PROGRAM#06:

Word Length Calculate

```
#include <iostream>  
using namespace std;  
  
int findLength (char word [])  
{  
    int count = 0;  
  
    while (word[count] != '\0')  
    {  
        count++;  
    }  
  
    return count;  
}
```



```
int main ()  
{  
    char word [100];  
  
    cin >> word;  
  
    cout << "Length of the word is: "  
        << findLength(word);  
  
    return 0;  
}
```